Case Study 2: Technical Presentation

STAT 946 – Case Studies in Data Science

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Section 1

Comparison and evaluation

Model Performance - Forecast

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Showdown plot saved as nhits_vs_lasso_plot.png!
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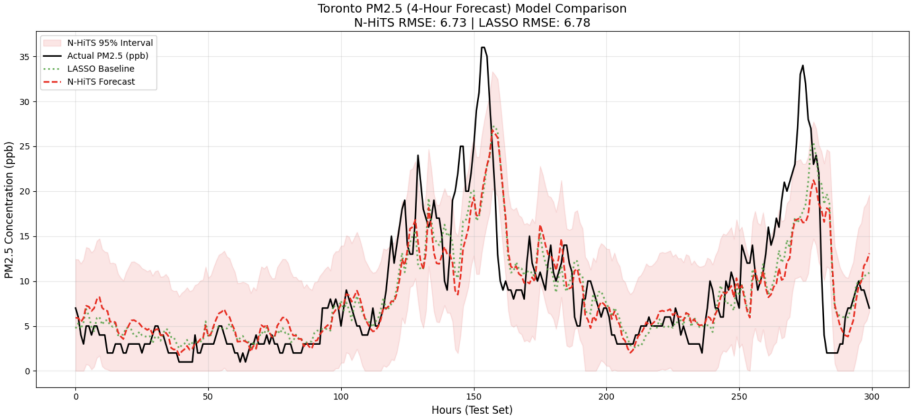

Figure 1: Toronto PM2.5 4-Hour Forecast Model Comparison

Model Performance - RMSE Comparison

Table 1: Model RMSE Comparison

Model	RMSE
LASSO	6.7843
N-HiTS	6.7347

- The LASSO and N-HiTS models produce very similar RMSE values, indicating that much of the predictive signal is captured by linear relationships and recent pollution history.
- The small improvement from N-HiTS suggests that nonlinear temporal structure exists but contributes only marginal additional predictive power for a 4-hour forecast horizon.

Overall Forecasting Accuracy and Effectiveness

- Typical hourly $PM_{2.5}$ levels in Ontario often range from ~5–30 μg/m^3, with occasional spikes above 50 μg/m^3 during pollution events. An RMSE of ~6.7 μg/m^3 therefore corresponds to an average prediction error of roughly 20–30% of typical concentration levels.
- This level of error is reasonable for short-term environmental forecasting, where pollution levels are highly variable and influenced by complex atmospheric processes.
- This forecasting system should be treated as an early-warning decision support tool, not a precise measurement system.
- Its value lies in anticipating pollution spikes early enough for operational adjustments, rather than predicting exact $PM_{2.5}$ concentrations.

Appendix: additional figures

Include:

TBD